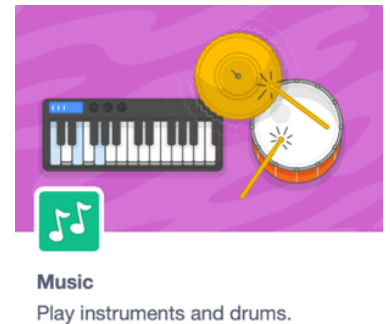


Scratch Looper



Register/login at <https://scratch.mit.edu>

Make rhythms by looping.

1) **Create** a new Project and add the Music extension.

*A **Beat** is a steady, recurring pulse in music that you can tap your foot to.*

***Tempo** is the speed at which sounds play, measured in BPM (beats per minute). The starting tempo in Scratch is **60 BPM**, which means 1 beat each second.*

1) Create code that **plays drums** for one beat, or 1 second. Start the code with the **green flag** and use a **forever** loop to make it repeat.

*Do you hear one beat per second?
This is your “rhythm section”*

*A **Bar** splits music into small, equal groups of beats. A lot of pop music has **4 beats** in every bar.*

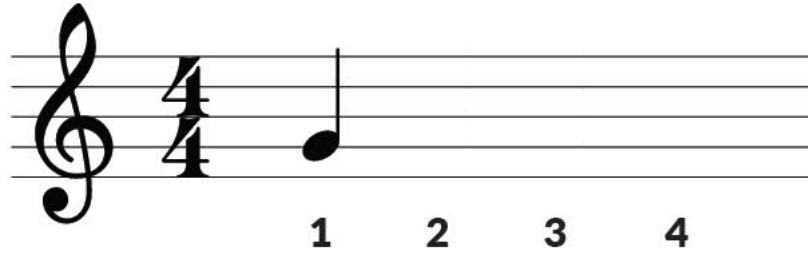


2) Add a new code block that also starts on the **green flag**. Make this code **repeat 4** beats to make a bar.

3) **Set an instrument** before you enter the loop and **play a note** for 1 beat, staying in time with the drums.

*Musicians call the first beat of the bar “**The One.**”*

- 4) You'll need to count the beats in the bar. Create a loop variable called **beats**.
- 5) To count 1, 2, 3, 4, set **beats** to 1 before you enter the loop and **change beat by 1** as the last thing in the loop.
- 6) Add an **if .. else** around the **note** so that it only plays on "The One." If it doesn't play (**else**) add a **rest** for 1 beat.



A **rest** is a silent delay.

- 7) We want the bar to repeat so put this loop and the setup of the loop variable inside another **forever** loop.

This is a **nested loop** as one loop is nested inside another.

- 8) Now, add some different notes on other beats in the bar. Add another **if .. else** where the **rest** is, and check for a different beat.
- 9) Don't forget to put the **rest** back in at the last **else** in the chain.

Have fun looping! You can always change the tempo by dragging a set tempo block onto your code space and just click on it.

Save your code with a good name.

File > Save now